

A photograph of a modern university building with large glass windows and a white facade, situated behind a field of tall grass and yellow wildflowers under a blue sky with scattered clouds. A semi-transparent grey box is overlaid on the left side of the image, containing text.

5CM512

Communication and Security Protocols

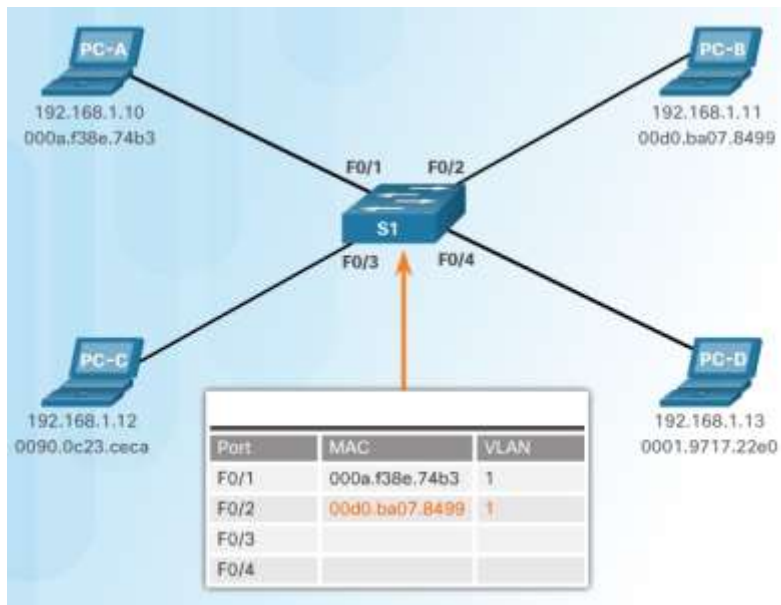
Week 10:
Security Attacks and
Countermeasures

Overview

- CAM tables attacks
- VLAN attacks
- DHCP attacks

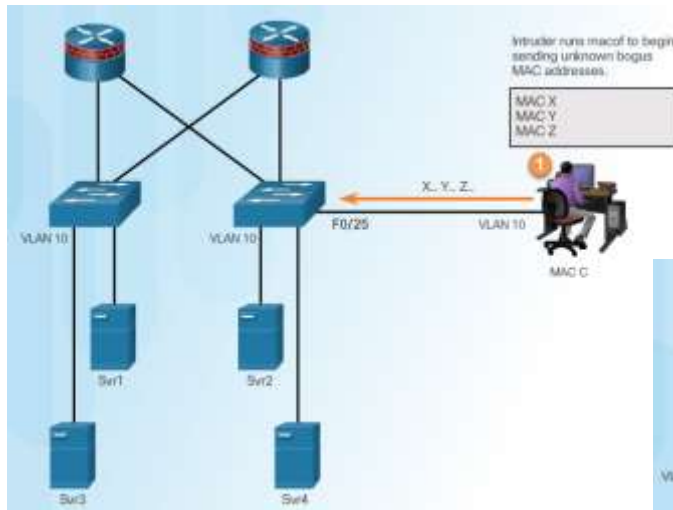
CAM TABLES ATTACKS

CAM Table Operation Example



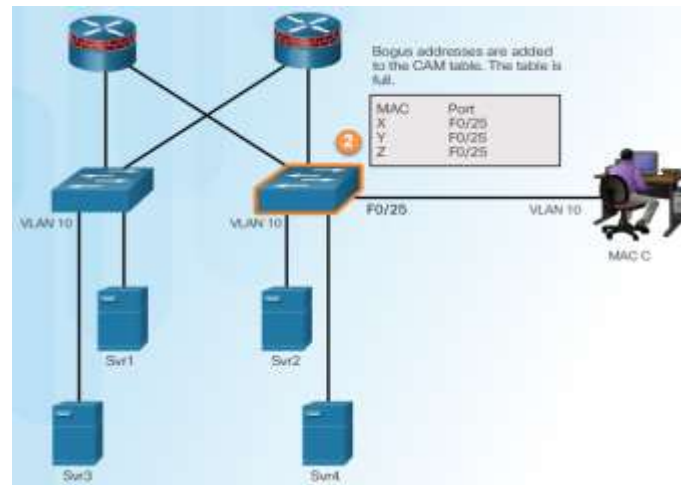
```
S1# show mac-address-table
Mac Address Table
-----
Vlan    Mac Address      Type        Ports
----    -
1       0001.9717.22e0   DYNAMIC     Fa0/4
1       000a.f38e.74b3   DYNAMIC     Fa0/1
1       0090.0c23.caca   DYNAMIC     Fa0/3
1       00d0.ba07.8499   DYNAMIC     Fa0/2
```

CAM Overflow Attack

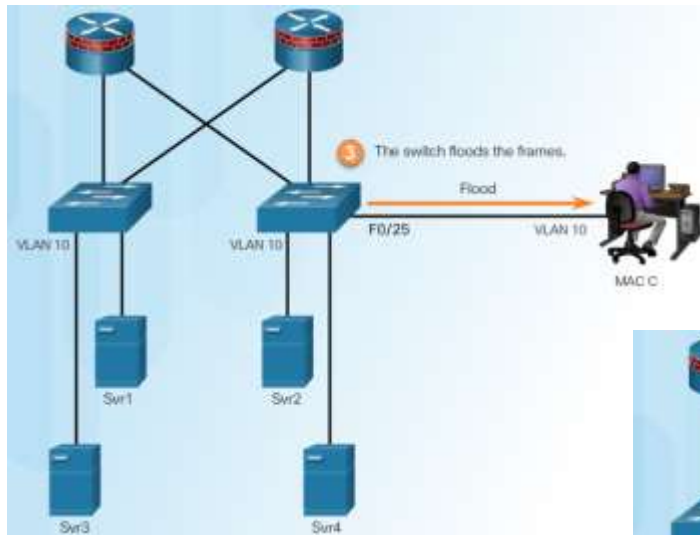


Fill CAM Table

Intruder Runs Attack Tool

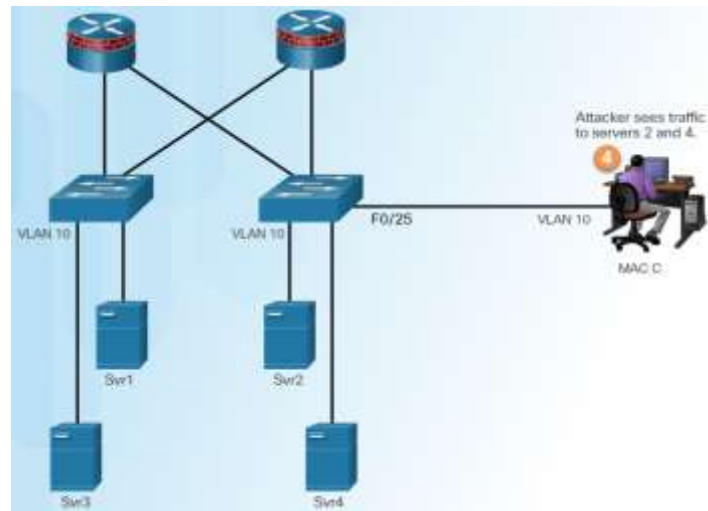


CAM Overflow Attack



Switch Floods All Traffic

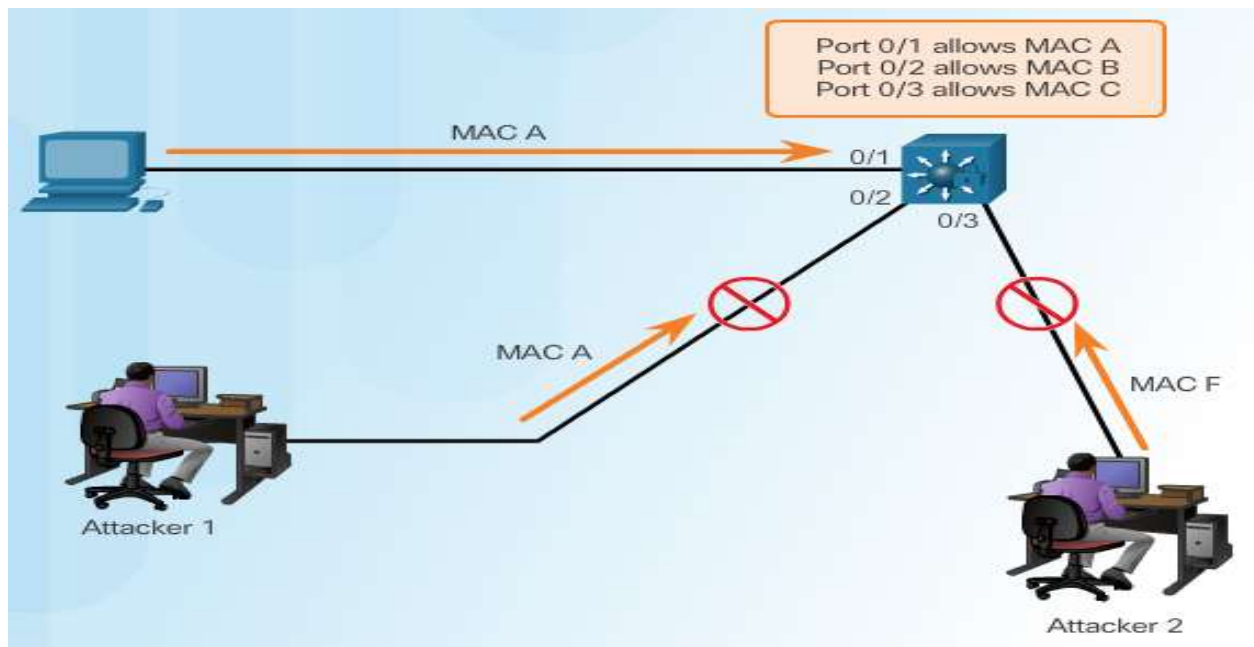
Attacker Captures Traffic



CAM Overflow Attack Tools

```
macof -i eth1
36:a1:48:63:81:70 15:26:8d:4d:28:f8 0.0.0.0.26413 > 0.0.0.0.49492: S 1094191437:1094191437(0) win 512
16:e8:8:0:4d:9c da:4d:bc:7c:ef:be 0.0.0.0.61376 > 0.0.0.0.47523: S 446486755:446486755(0) win 512
18:2a:de:56:38:71 33:af:9b:5:a6:97 0.0.0.0.20086 > 0.0.0.0.6728: S 105051945:105051945(0) win 512
e7:5c:97:42:ec:1 83:73:1a:32:20:93 0.0.0.0.45282 > 0.0.0.0.24898: S 1838062028:1838062028(0) win 512
62:69:d3:1c:79:ef 80:13:35:4:cb:d0 0.0.0.0.11587 > 0.0.0.0.7723: S 1792413296:1792413296(0) win 512
c5:a:b7:3e:3c:7a 3a:ee:c0:23:4a:fe 0.0.0.0.19784 > 0.0.0.0.57433: S 1018924173:1018924173(0) win 512
88:43:ee:51:c7:68 b4:8d:ec:3e:14:bb 0.0.0.0.283 > 0.0.0.0.11466: S 727776406:727776406(0) win 512
b8:7a:7a:2d:2c:ae c2:fa:2d:7d:e7:bf 0.0.0.0.32650 > 0.0.0.0.11324: S 605528173:605528173(0) win 512
e0:d8:1e:74:1:e 57:98:b6:5a:fa:de 0.0.0.0.36346 > 0.0.0.0.55700: S 2128143986:2128143986(0) win 512
```

Countermeasure for CAM Overflow Attacks



Port Security

Port Security Options

```
S1(config)# interface f0/1
S1(config-if)# switchport port-security ?
    aging      Port-security aging commands
    mac-address Secure mac address
    maximum    Max secure addresses
    violation   Security violation mode
    <cr>

S1(config-if)# switchport port-security
```

Enabling Port Security

```
S1(config)# interface f0/1
S1(config-if)# switchport port-security
Command rejected: FastEthernet0/1 is a dynamic port.
S1(config-if)# switchport mode access
S1(config-if)# switchport port-security
S1(config-if)# end
S1#
```

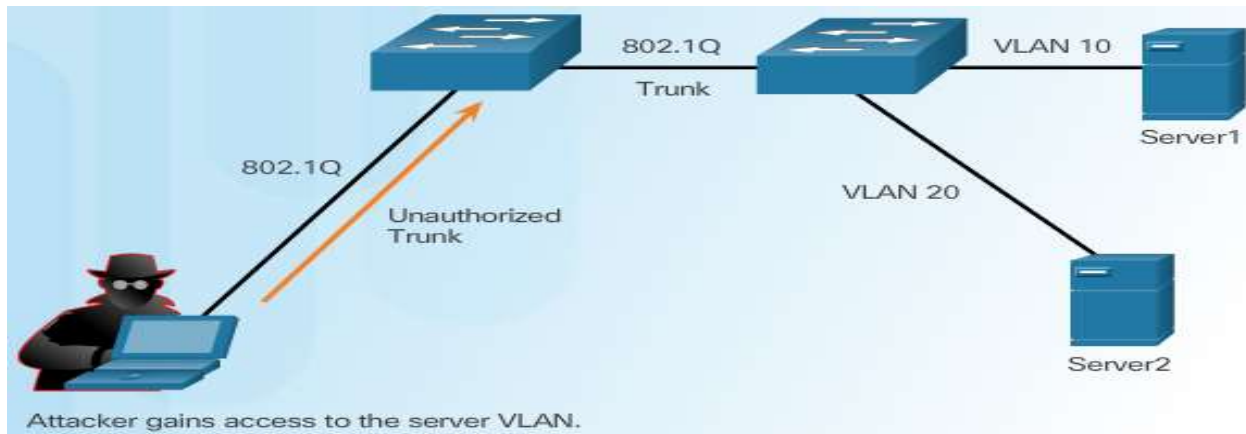
```
S1# show port-security interface f0/1
Port Security          : Enabled
Port Status            : Secure-shutdown
Violation Mode         : Shutdown
Aging Time             : 0 mins
Aging Type             : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses  : 1
Total MAC Addresses    : 0
Configured MAC Addresses : 0
Sticky MAC Addresses   : 0
Last Source Address:Vlan : 0000.0000.0000:0
Security Violation Count : 0
S1#
```

Security Violation Modes

Violation Mode	Forwards Traffic	Sends Syslog Message	Increases Violation Counter	Shuts Down Port
Protect	No	No	No	No
Restrict	No	Yes	Yes	No
Shutdown	No	Yes	Yes	Yes

VLAN ATTACKS

VLAN Hopping Attacks: Switch Spoofing

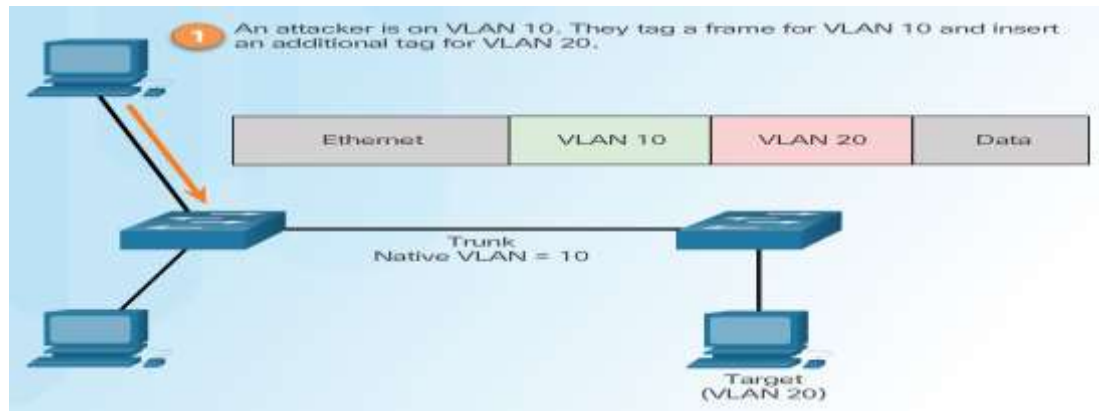


Dynamic
Trunking
Protocol (DTP)
Modes
Combination

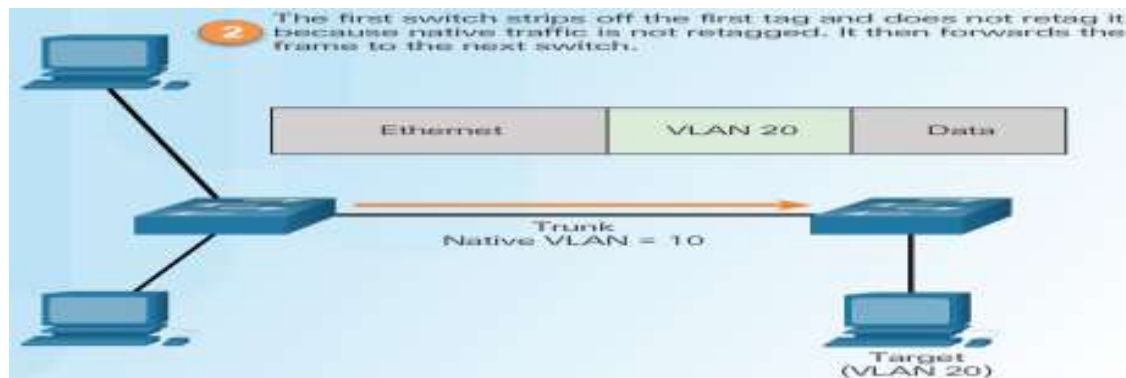
	Dynamic Auto	Dynamic Desirable	Trunk	Access
Dynamic Auto	Access	Trunk	Trunk	Access
Dynamic Desirable	Trunk	Trunk	Trunk	Access
Trunk	Trunk	Trunk	Trunk	Limited Connectivity
Access	Access	Access	Limited Connectivity	Access

VLAN Hopping: Double-Tagging Attack I

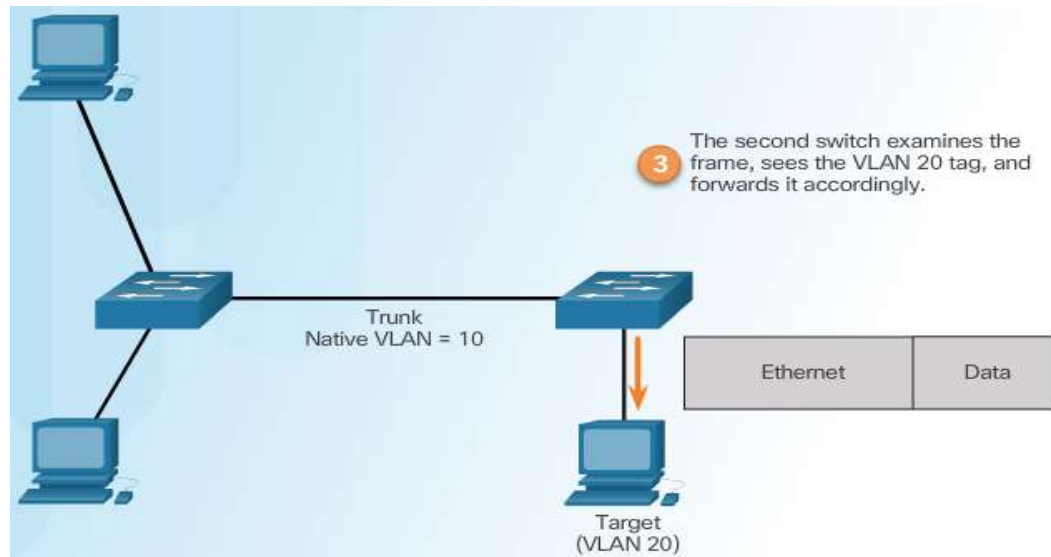
Step 1 –
Double
Tagging
Attack



Step 2 –
Double
Tagging
Attack

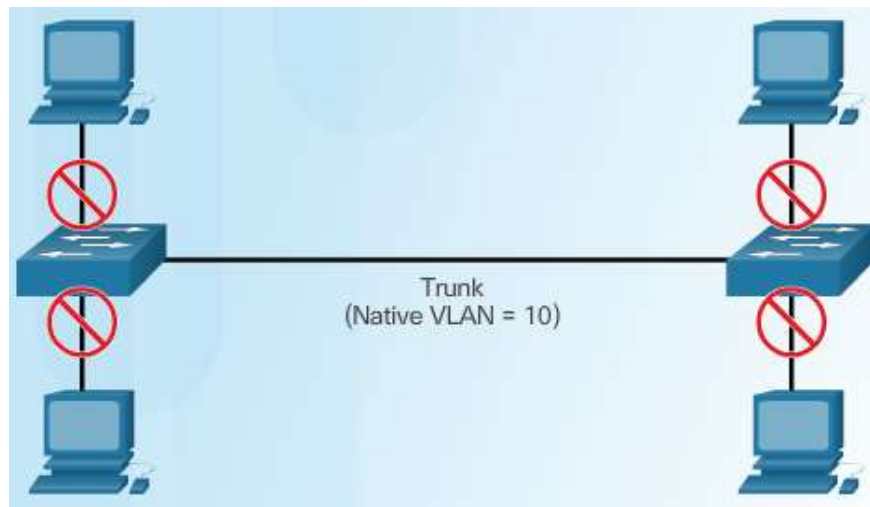


VLAN Hopping: Double-Tagging Attack II



Step 3 – Double Tagging Attack

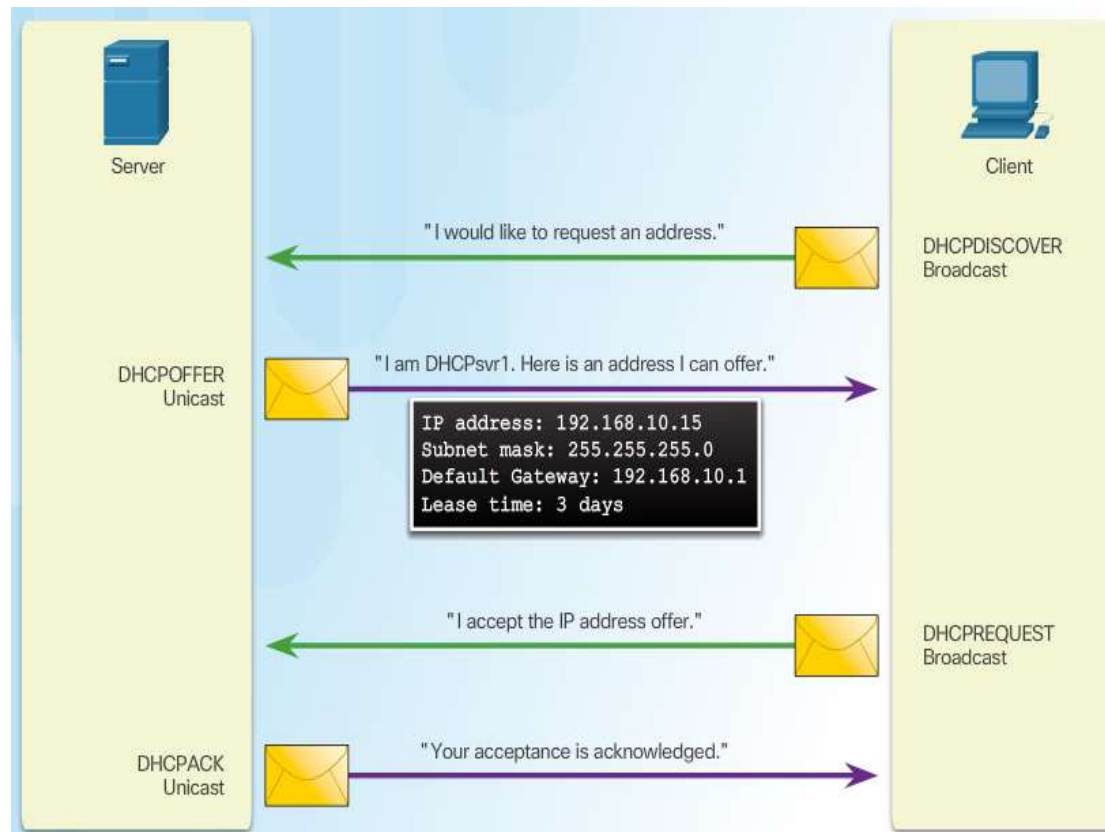
Mitigating VLAN Hopping Attacks



- Disable trunking on all access ports.
- Disable auto trunking and manually enable trunking.
- Be sure that the native VLAN is used only for trunk lines.

DHCP ATTACKS

DHCP Message Exchange

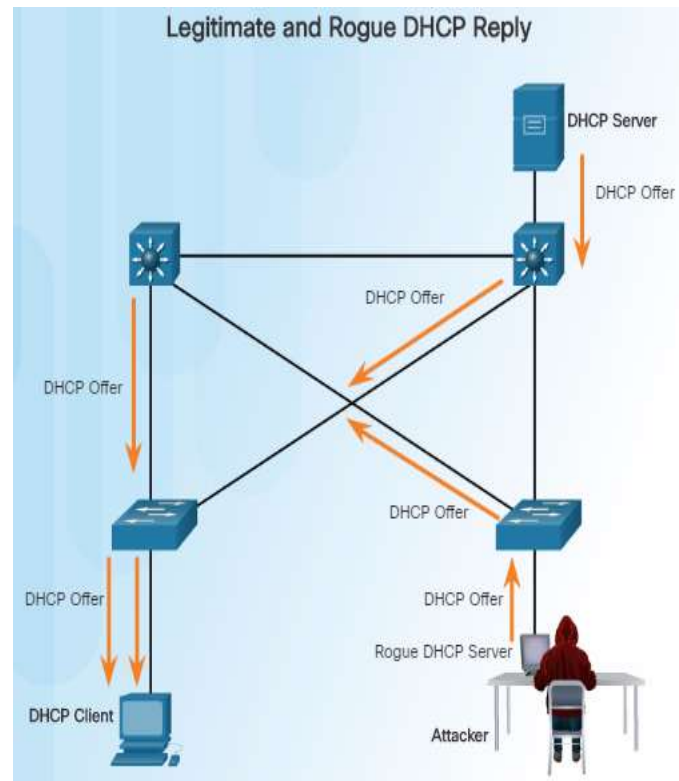
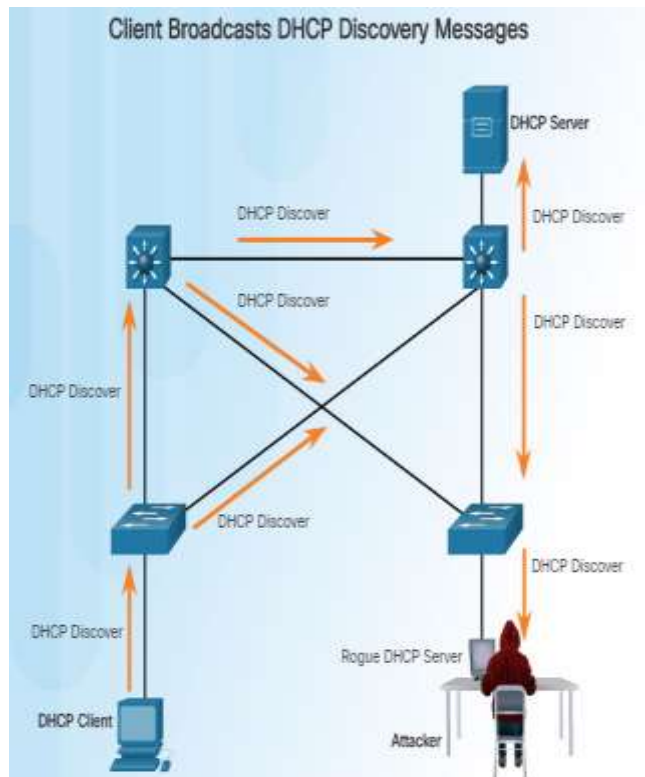


DHCP Spoofing Attack (cont.)

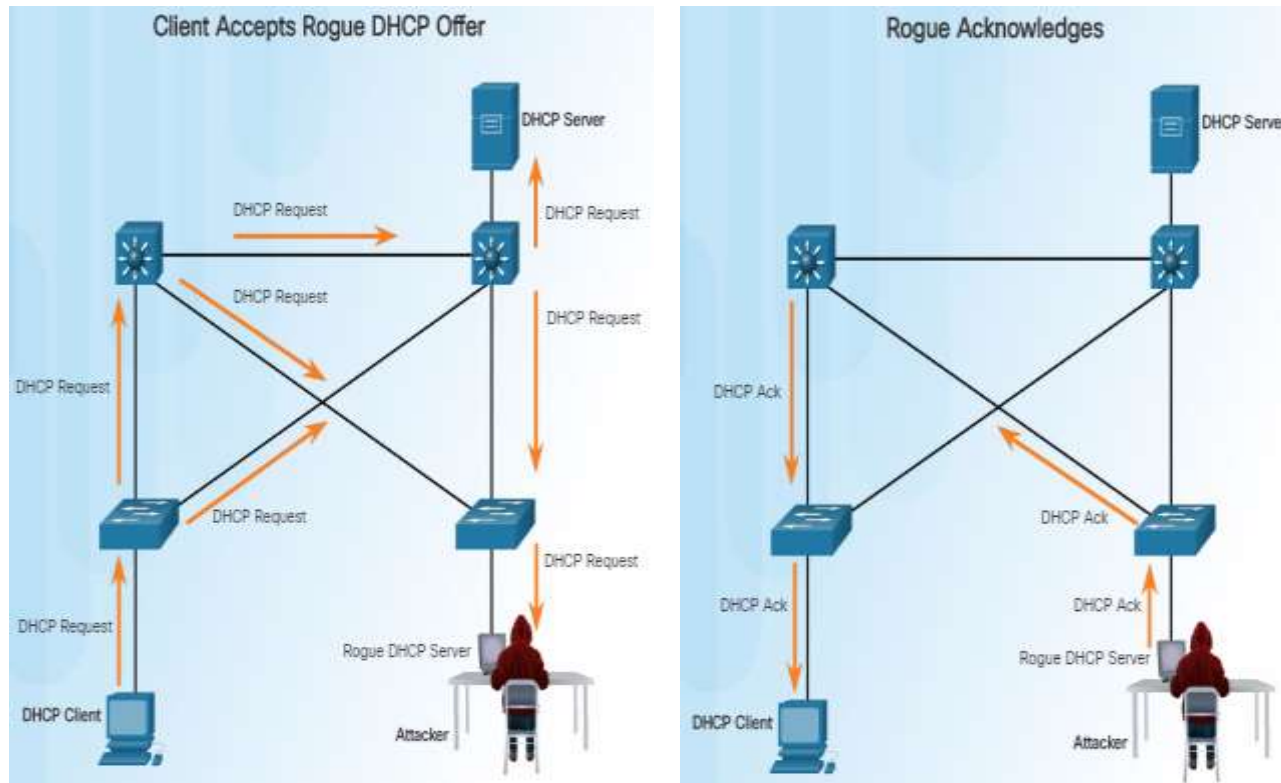
A DHCP spoofing attack occurs when a rogue DHCP server is connected to the network and provides false IP configuration parameters to legitimate clients. A rogue server can provide a variety of misleading information:

- **Wrong default gateway** - Attacker provides an invalid gateway or the IP address of its host to create a man-in-the-middle attack. This may go entirely undetected as the intruder intercepts the data flow through the network.
- **Wrong DNS server** - Attacker provides an incorrect DNS server address pointing the user to a nefarious website.
- **Wrong IP address** - Attacker provides an invalid default gateway IP address and creates a DoS attack on the DHCP client.

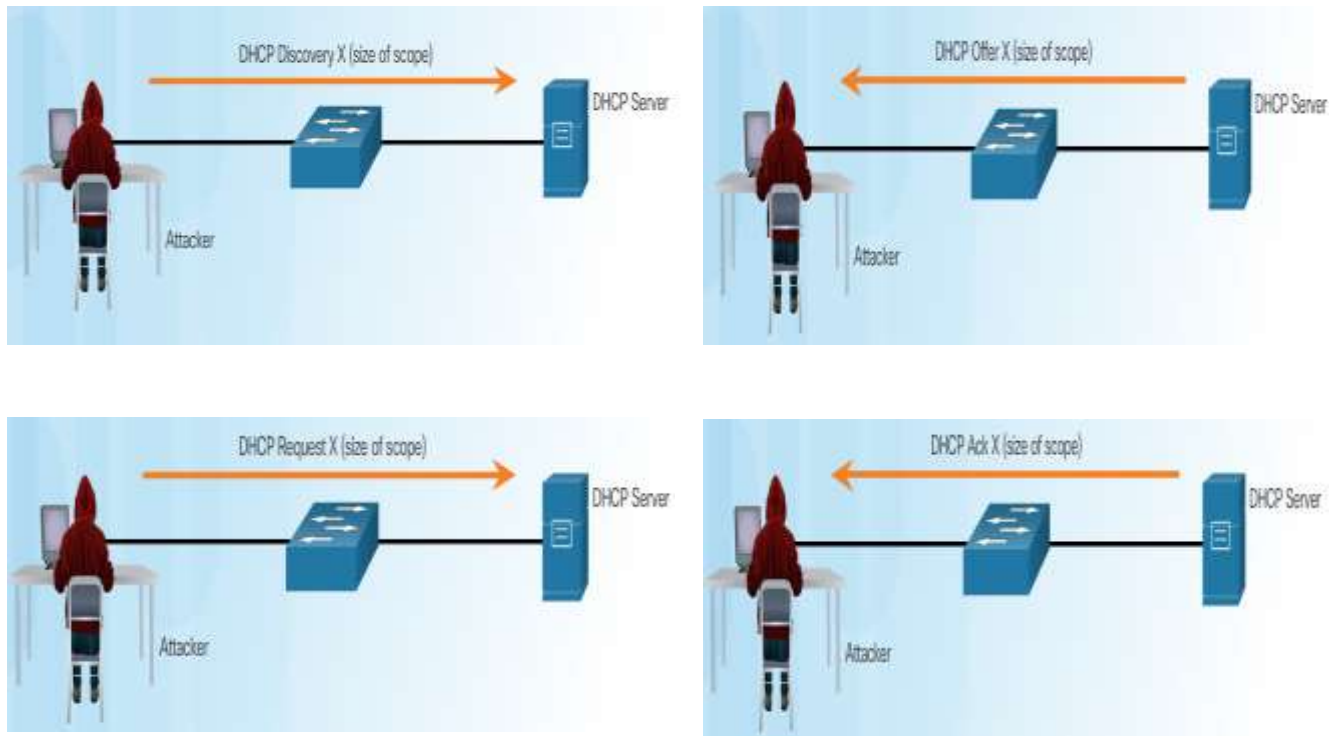
DHCP Spoofing Attack (cont.)



DHCP Spoofing Attack



DHCP Starvation Attack

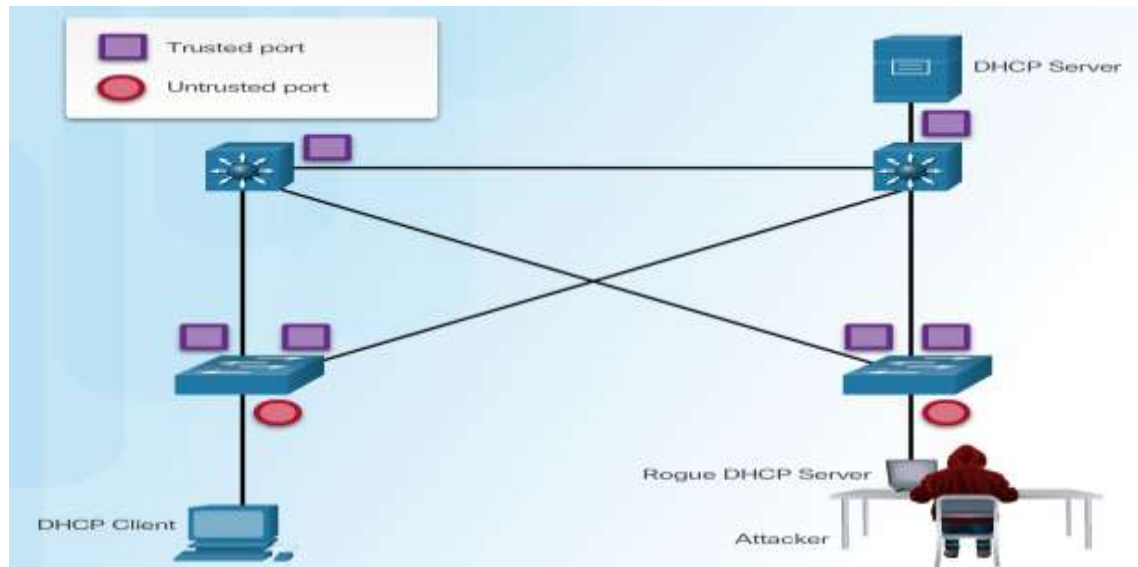


The goal of this attack is to create a DoS for connecting clients.

DHCP Snooping

- DHCP spoofing attacks can be mitigated using DHCP snooping on trusted ports.
- DHCP snooping also helps mitigate against DHCP starvation attacks by rate limiting the number of DHCP discovery messages that an untrusted port can receive.
- DHCP snooping builds and maintains a DHCP snooping binding database that the switch can use to filter DHCP messages from untrusted sources.
- The DHCP snooping binding table includes the client MAC address, IP address, DHCP lease time, binding type, VLAN number, and interface information on each untrusted switchport or interface.
- The switch will deny packets containing specific information:
 - Unauthorized DHCP server messages from an untrusted port
 - Unauthorized DHCP client messages not adhering to the snooping binding table or rate limits
 - DHCP relay-agent packets that include option-82 information on an untrusted port

Configuring DHCP Snooping



- Trusted DHCP ports - Only ports connecting to upstream DHCP servers should be trusted. These ports that are expected to reply with DHCP offer and DHCP Ack messages.
- Untrusted ports - These ports connect to hosts that should not be providing DHCP server messages.

Configuring DHCP Snooping Example

DHCP Snooping Reference Topology



Configuring a Maximum Number of MAC Addresses

```
S1(config)# ip dhcp snooping
S1(config)#
S1(config)# interface f0/1
S1(config-if)# ip dhcp snooping trust
S1(config-if)# exit
S1(config)#
S1(config)# interface range f0/5 - 24
S1(config-if-range)# ip dhcp snooping limit rate 6
S1(config-if-range)# exit
S1(config)#
S1(config)# ip dhcp snooping vlan 5,10,50-52
S1(config)#
```

Any QUESTIONS?

